Topics of today's class

Last class:

- Chapter 1
  - Introduction to Macroeconomics
  - Recent and Current macroeconomic Events

Today's class:

- Chapter 2
  - Measuring GDP using three approaches
  - Problem in measuring GDP

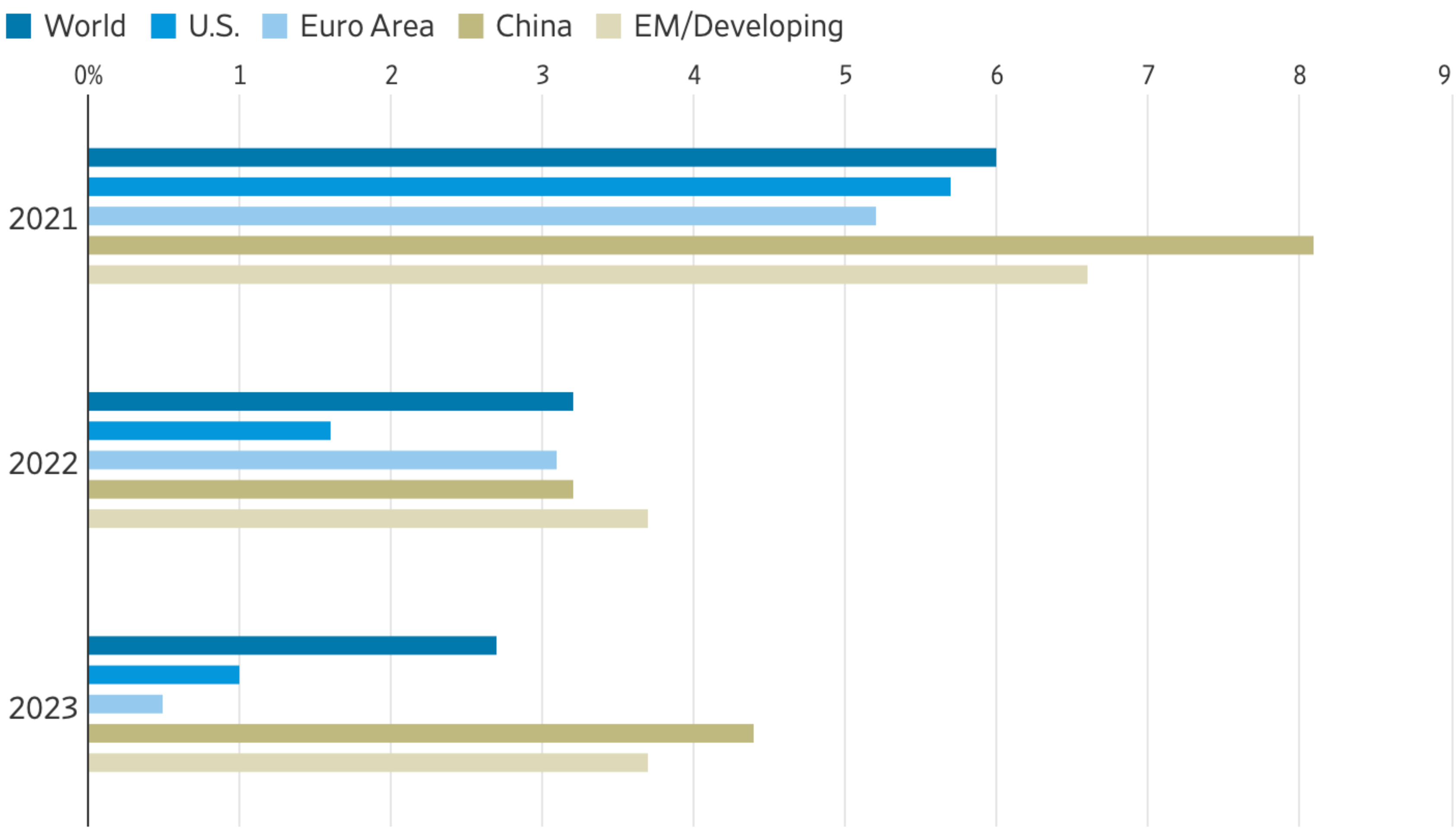
Chapter 2

Measurement

Why Study Measurement?

- *"Data, data, data, I cannot make bricks without clay" – Sherlock Holmes.*
- Macroeconomics produces hypothesis about aggregate policy.
- Without a credible measurement of these quantities, it is impossible to test the validity of these theories.
- Economic data are crucial for understanding the state of the economy.

Measuring GDP: The National Income and Product Accounts (NIPA)



Note: 2022 and 2023 numbers are forecasts. 2021 numbers are actual growth.
Source: International Monetary Fund

Measuring GDP: The National Income and Product Accounts (NIPA)

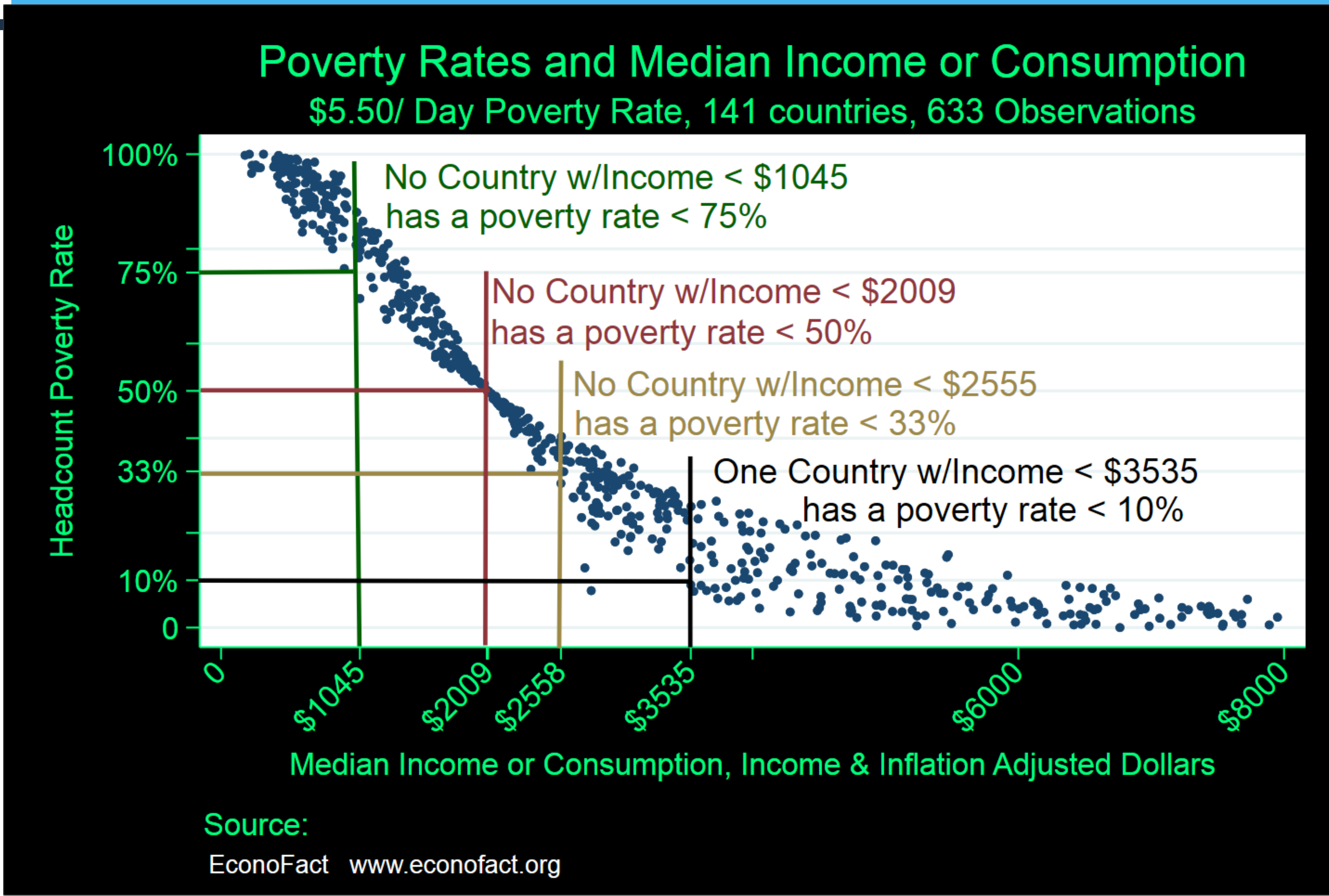
- Nominal GDP is the dollar value of final output produced during a given period of time within the borders of the U.S.
- GDP intended to summarize how much economic activity is going during a specific period.
- The NIPA accounts are a conceptual framework for organizing data on the production of goods and services and on the incomes received by factors of production.
  - published by the U.S. Department of Commerce

Why should we care about GDP?

"Yet ***the gross national product*** does not allow for the health of our children, the quality of their education, or the joy of their play. It does not include the beauty of our poetry or the strength of our marriages; the intelligence of our public debate or the integrity of our public officials. It measures neither our wit nor our courage; neither our wisdom nor our learning; neither our compassion nor our devotion to our country; it measures everything, in short, except that which makes life worthwhile. And it ***tells us everything about America except why we are proud that we are Americans***."

Robert F. Kennedy
Speech at the University of Kansas
March 18, 1968

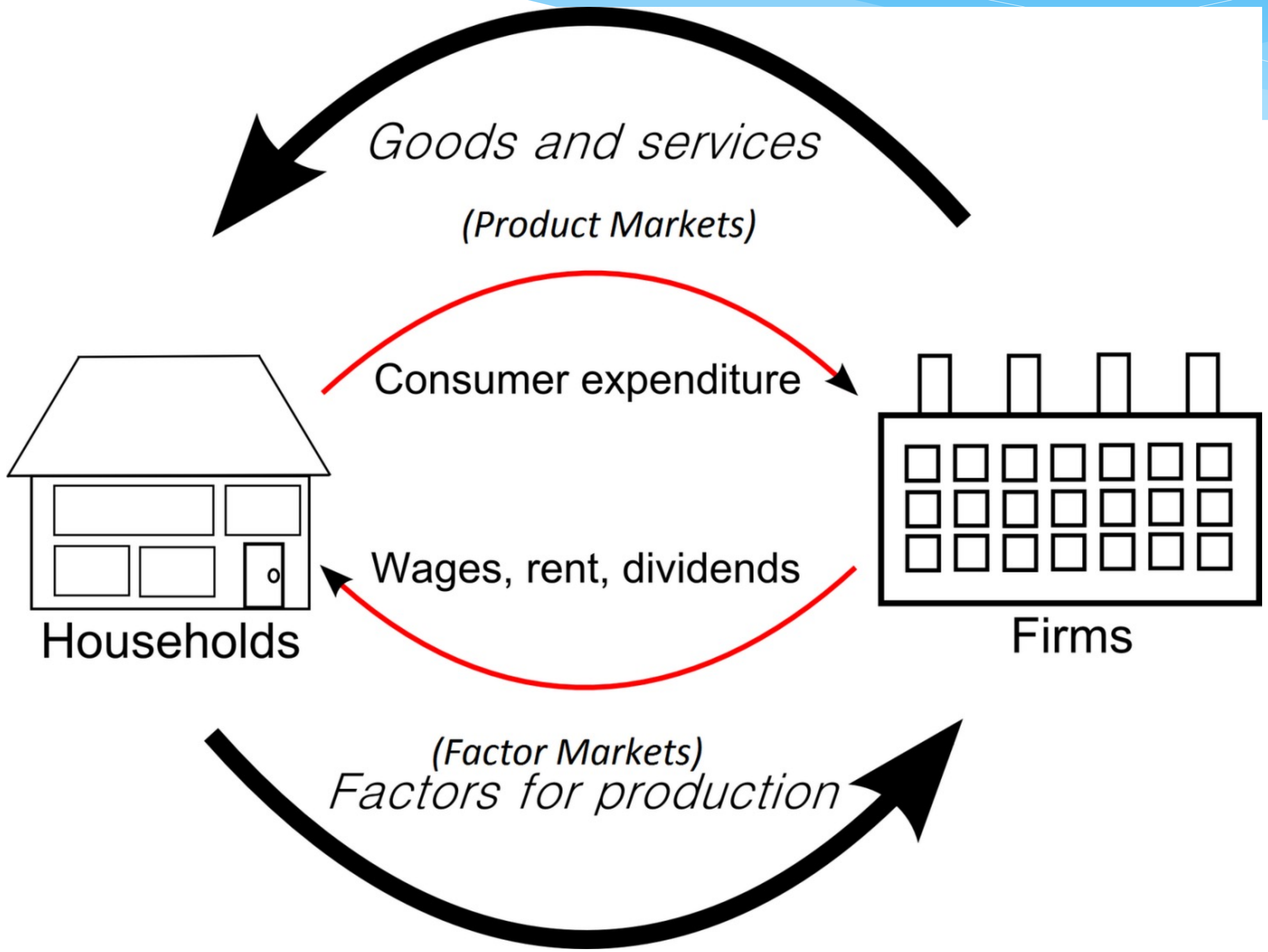
GDP and absolute poverty



NIPA

- Three equivalent ways to think of GDP:
  - The product approach: total *production* of everyone in the economy
  - The expenditure approach: total *expenditure* of everyone in the economy
  - The income approach: total *income* of everyone in the economy
- They must all give the same results. (excluding measurement errors)

The Simple Circular Flow Diagram



National Income Accounting Example

- Fictional Island Economy: a coconut producer, a restaurant, consumers, and a government.
- ***Coconut Producer***: owns all the coconut trees, harvests the coconuts that grow on the trees.
- ***Restaurant***: buys coconuts from the coconut producer and serves "coconut soup" to its client.
- ***Consumers***: buy coconuts or drinks and work.
- ***Government***: provides national defense.

National Income Accounting Example

- ***Coconut producer***: produces 10 million coconut, each sold at $2. 6 million coconuts go to the restaurant, remaining 4 million go to the consumers.
- ***Restaurant***: sells $30 million meals.
- ***Government***: provides protection from attacks from other islands. It collects taxes ($4.5 million from firms, 1 million from consumers).
- ***Consumers***: work for the business and government, receive interest, pay taxes.

Table 2.1
Coconut Producer

Table 2.1 Coconut Producer

Total Revenue	\$20 million
Wages	\$5 million
Interest on Loan	\$0.5 million
Taxes	\$1.5 million

Table 2.2
Restaurant

Table 2.2 Restaurant

Total Revenue	$30 million
Cost of Coconuts	$12 million
Wages	$4 million
Taxes	$3 million

Table 2.3
After-Tax Profits

Table 2.3 After-Tax Profits

Coconut Producer	$13 million
Restaurant	$11 million

Table 2.4
Government

Table 2.4	Government
Tax Revenue	$5.5 million
Wages	$5.5 million

Table 2.5
Consumers

Table 2.5 Consumers

Wage Income	$14.5 million
Interest Income	$0.5 million
Taxes	$1 million
Profits Distributed to Producers	$24 million

Product Approach

- Also called the Value-added approach.
- GDP = sum of value-added to goods and services across all productive units in the economy
  - ***Value Added*** is the increase in the value of goods as a result of the production process
  - ***Value-added*** = value of production – value of intermediate goods
  - an ***intermediate good*** is a good that is produced and then used as an input to another production process
  - Which good is an intermediate good in this economy?

Product Approach

- Government production is not sold at market prices
- What is the Value added of the government?

  = cost of the inputs to production (standard practice)

Table 2.6
GDP Using the Product Approach

Table 2.6 GDP Using the Product Approach	
Value added - coconuts	$20 million
Value added - restaurant food	$18 million
Value added - government	$5.5 million
GDP	$43.5 million

Expenditure Approach

- Measure the amount that people spend.

$$Y = C + I + G + NX$$

  - Y: Nominal GDP
  - C: Consumption Expenditures: durable goods, non-durables, and services.
  - I: Investment Expenditures
  - G: Government Expenditures: Sum of federal, state, and local purchases of goods and services, and government investment.
  - $NX = X - M$: net exports: total export of US goods and services minus total imports into the US.

Table 2.7
GDP Using the Expenditure Approach

Table 2.7 GDP Using the Expenditure Approach

Consumption	$38 million
Investment	0
Government Expenditures	$5.5 million
Net Exports	0
GDP	$43.5 million

Income Approach

- Measure income that people received.
- GDP = Wage Income

  + After-Tax Profits

  + Interest Income

  + Taxes

Table 2.8
GDP Using the Income Approach

Table 2.8 GDP Using the Income Approach

Wage Income	$14.5 million
After-tax profits	$24 million
Interest Income	$0.5 million
Taxes	$4.5 million
GDP	$43.5 million

Magic?

Why do they all give the same result?

- total output, or value-added, is ultimately sold thus shows up as an expenditure
- what is spend on all output produced is income for someone in the economy.

Table 2.9
Components of Nominal GDP

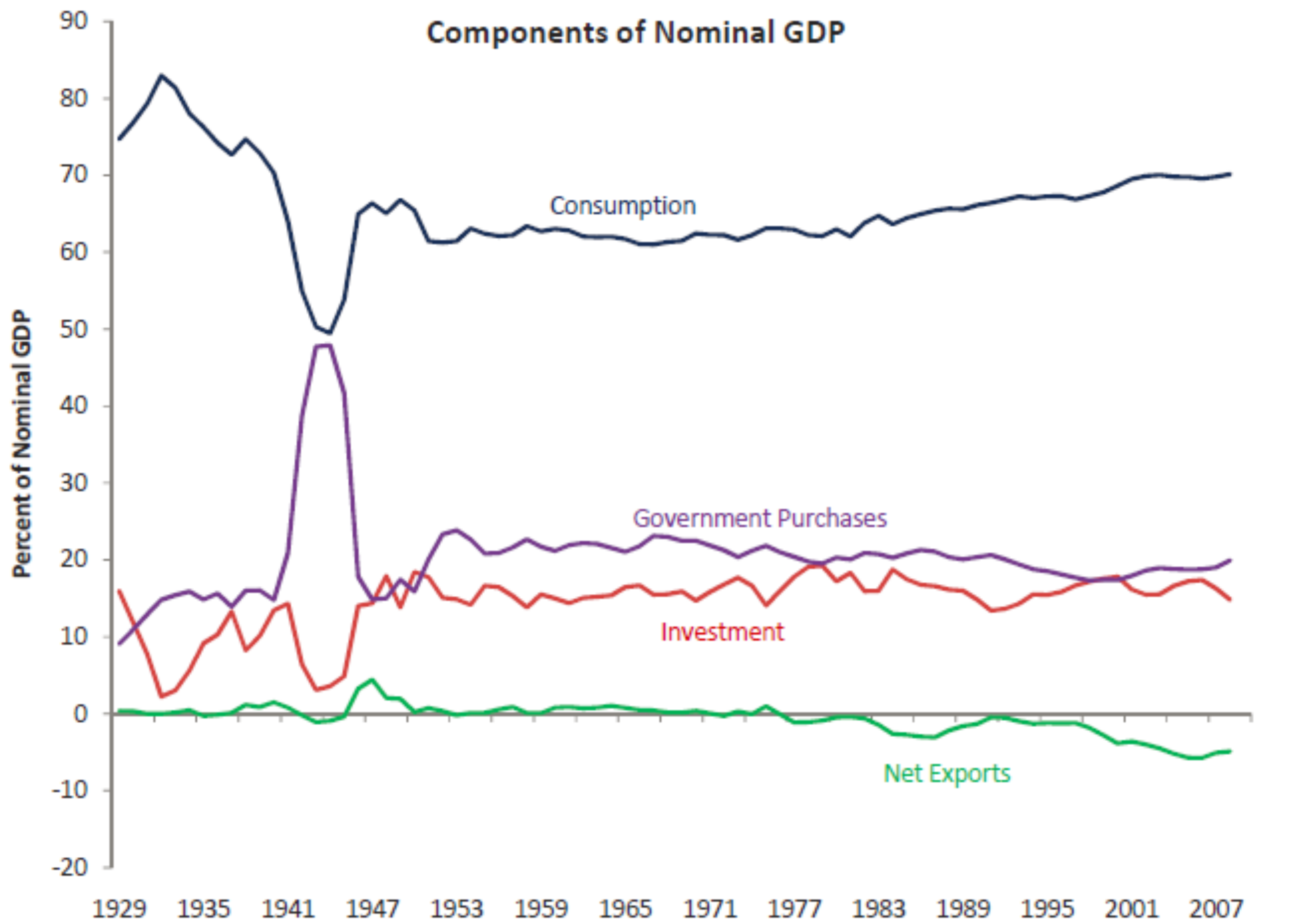


Table 2.9
GDP of the United States for 1961 & 2011

	1961	% of GDP	2011	% of GDP
Gross domestic product	544.8	100	15075.7	100
Personal consumption expenditures	342.2	62.81	10729	71.17
Goods	178.8	32.82	3624.8	24.04
Durable goods	44.2	8.113	1146.4	7.604
Nondurable goods	134.6	24.71	2478.4	16.44
Services	163.4	29.99	7104.2	47.12
Gross private domestic investment	78.2	14.35	1854.9	12.3
Fixed investment	75.2	13.8	1818.3	12.06
Nonresidential	48.8	8.957	1479.6	9.814
Residential	26.4	4.846	338.7	2.247
Change in private inventories	3	0.551	36.6	0.243
Net exports of goods and services	4.9	0.899	-568.1	-3.768
Exports	27.6	5.066	2094.2	13.89
Imports	22.7	4.167	2662.3	17.66
Government consumption expenditures and gross investment	119.5	21.93	3059.8	20.3
Federal	67.9	12.46	1222.1	8.106
National defense	56.5	10.37	820.8	5.445
Nondefense	11.4	2.093	401.3	2.662
State and local	51.6	9.471	1837.7	12.19

Add in Inventory Investment

- Suppose now the coconut producer produces 13 million instead 10 million, but only sell out 10 million. What is GDP now?

Answer: $43.5+3*$2 = $49.5 million.

Add in International Trade

- Suppose now the restaurant imports additional 2 million coconuts from other islands at $2 each. What is GDP now?

  Answer: $43.5-2*$2 = $39.5 million.

Problems in Measuring GDP

GDP per capita partly captures welfare. BUT:

- Domestic Production
  - Hiring a professional cleaner (in GDP) VS cleaning your apartment yourself (not in GDP).
  - increase in female labor force participation since WWII: may over-estimate GDP growth.
- Underground economy
  - paying a babysitter in cash.
  - illegal drugs, criminality activity.
- Government production is difficult to measure as the output is typically not sold in the market.
- Inequality.


Advil

What is GNP?

- GNP = GDP + NFP

  NFP: Net Factor Payments

  = Factor payments to US citizens

  - Factor payments to foreigners
- GNP: output produced by domestically owned factors
  - unlike GDP, GNP is ***not*** geographic concept -> GDP is based on geographic borders of a country.
  - GNP adds together income paid to nationally owned factors of production regardless of where in the world they are located
- Difference between GDP and GNP for US is small (0.2%), larger for countries with many citizens working abroad

What is GNP?

Example:

- if a Volvo Cars plant in Ohio has Chinese Owners but American workers:
  - its profits are in the GDP but not in the GNP of the U.S.
  - its wage bill is in the GDP and in the GNP of the U.S.
- if a Google office in China is owned by US residents but Chinese engineers operate the branch:
  - its profits are in the GNP but not in the GDP of the U.S.
  - its wage bill is not in the GDP nor in the GNP of the U.S.
- My income is paid by the State of Nebraska. But I am a Chinese citizen. It is counted in the US GDP but not in the US GNP, in China GNP but not in China GDP.

Question

Assume an economy with a **coal producer**, a **steel producer**, and some **consumers** (there is no government).

In a given year, the coal producer produced 15 million tons of **coal** and sells it for $5 per ton. The coal producer pays $50 million in wages to consumers.

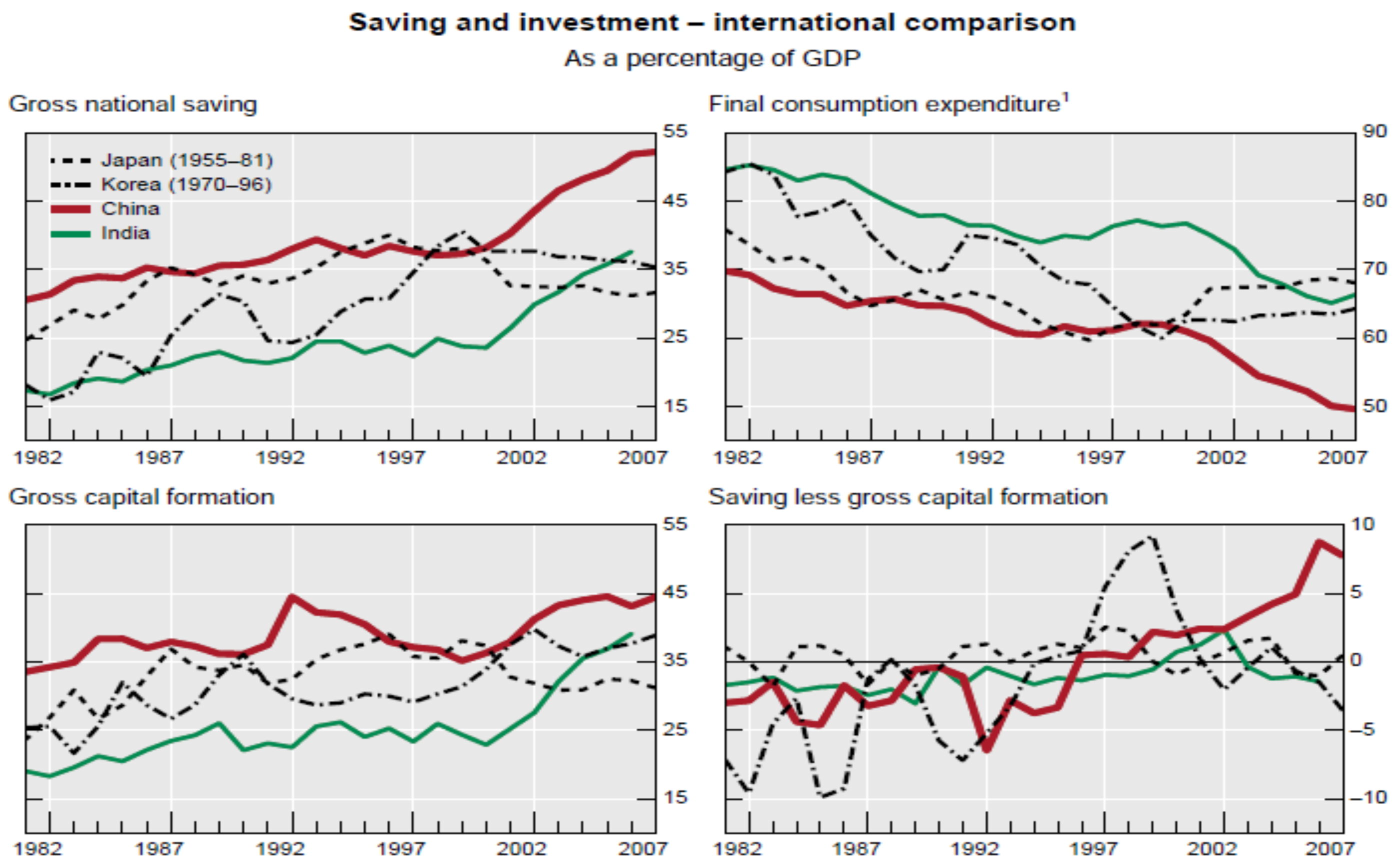
The steel producer uses 25 million tons of coal as an input into **steel** production, all purchased at $5 per ton. Of this, 15 million tons of coal comes from the **domestic** coal producer and 10 million tons of coal is **imported**. The steel producer produces 10 million tons of steel and sells it for $20 per ton.

Domestic consumers buy 8 million tons of steel, and 2 million tons are **exported**. The steel producer pays consumers $40 million in wages. All profits made by domestic producers are distributed to domestic consumers.

(a) Determine GDP using three approaches.

(b) Determine GNP in the case where the coal producer is owned by foreigners.

Case study: gender ratio and saving rate



[1] Including both private and government final consumption expenditure.

Source: Ma and Yi (2010), Bank for International Settlements

Case study: gender ratio and saving rate

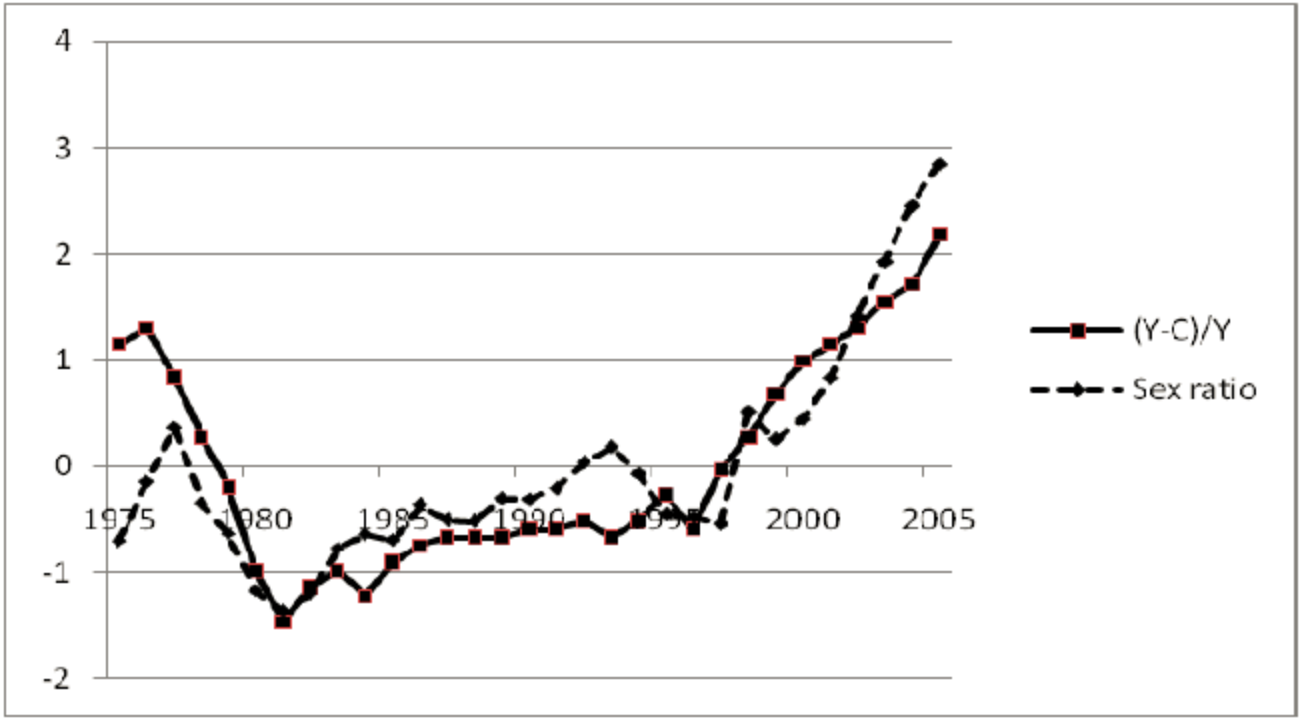


Figure 1: Sex Ratios and Saving Rates

Source: Wei and Zhang (2011), Journal of Political Economy